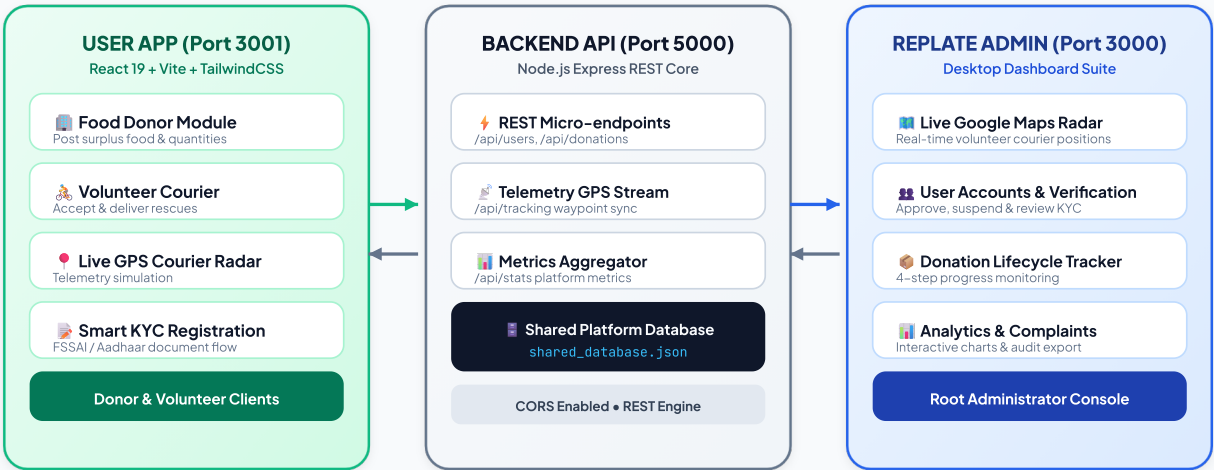




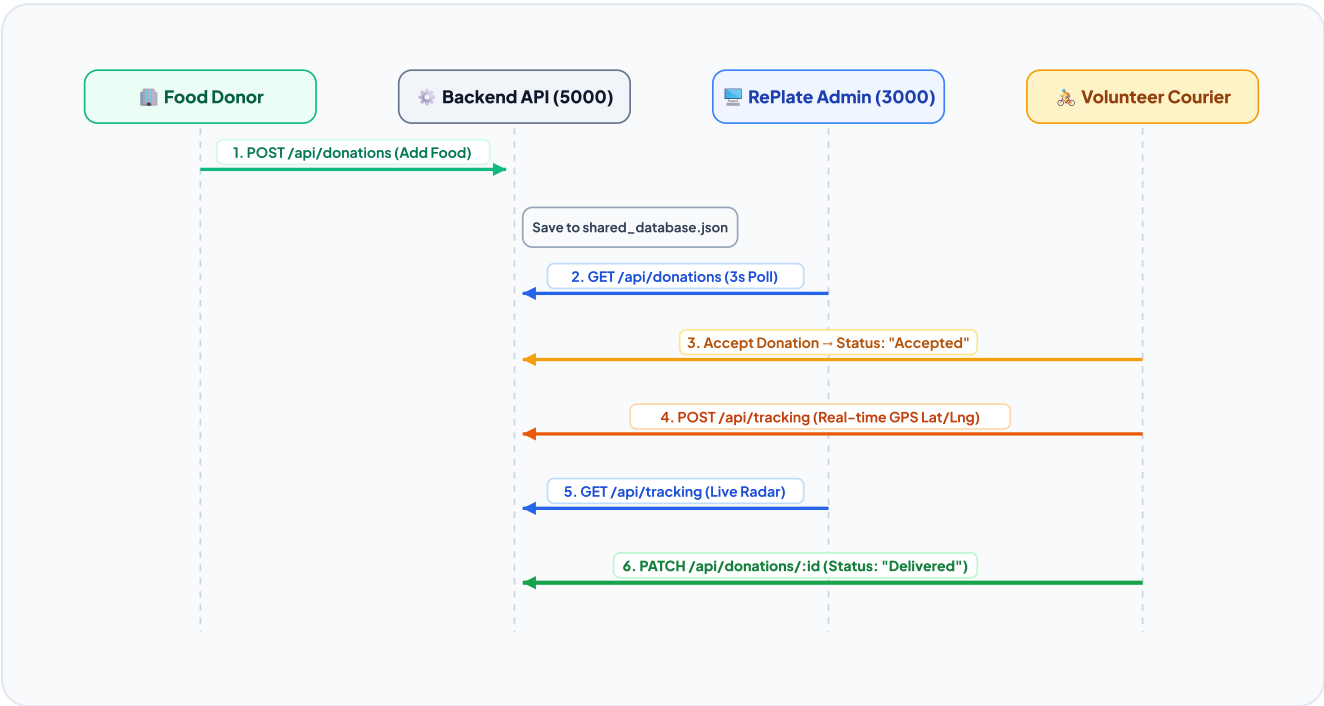
1. System Architecture Overview

The RePlate food rescue ecosystem operates as a coordinated 3-tier decentralized client-server architecture. Donors, volunteers, and centralized super-admins interact seamlessly through real-time REST API micro-endpoints and synchronized telemetry states.



2. Operational Working Flow Lifecycle

The sequence protocol traces every stage of the surplus food recovery process—from donor dispatch to volunteer courier drop-off, synchronized with backend persistence and live admin telemetry:



MONITORING LIFECYCLE STAGE	ADMIN VISIBILITY	VOLUNTEER TELEMETRY	BACKEND EVENT
1. Food Uploaded	Visible Instantly	Broadcasted to open feeds	POST /api/donations
2. Courier Assigned	Visible Instantly	Assigned courier name locked	PATCH /api/donations/:id
3. Pickup & Transit	Live Radar Active	Waypoints stream every 2s	POST /api/tracking
4. Verified Drop-off	Archived in Rescues	Delivered badge triggered	Milestone incremented (+Meals)

✓ **3. Functional Verification Matrix**

A. Food Donor & Volunteer Mobile App (Port 3001)

CAPABILITY / WORKFLOW	AUDIT STATUS	OPERATIONAL DETAILS
Donor 1-Click Login	✓ WORKING	Pre-filled credentials (<code>donor@foodrescue.org</code>) & 1-click quick-login button
Volunteer Quick Login	✓ WORKING	Courier persona quick-launch with persistent local session storage
Add Food Donation	✓ WORKING	Food category, meal count, preparation timestamp, shelf life, and GPS pickup coordinates
Multi-step Registration	✓ WORKING	Step 1 (Personal), Step 2 (FSSAI license / Aadhaar), Step 3 (Terms & auto-submit)
Volunteer Map & Route	✓ WORKING	Curved route visualization to Anbu Illam, Chidambaram with live waypoint pings

B. RePlate Central Admin Panel (Port 3000)

ADMIN CONTROL TOWER FEATURE	AUDIT STATUS	OPERATIONAL DETAILS
Super Admin Authentication	✓ WORKING	Restricted credentials (<code>admin@foodsave.org</code> / <code>admin123</code>) with 1-click auto-fill
Desktop Website Layout	✓ UPGRADED	Full desktop width (no mobile frame), modern sidebar navigation, top header breadcrumbs
Live Google Maps Telemetry	✓ WORKING	Real Google Maps API integration with active volunteer courier marker tracking
Volunteer Radar Console	✓ WORKING	Live satellite telemetry feed showing real-time coordinates, last ping, and route destination
User Verification & Moderation	✓ WORKING	Verify pending NGOs/volunteers, suspend/block flagged bad actors with sync to backend
Executive Analytics Export	✓ WORKING	Consolidated CSV/Excel & printable HTML report generator for platform auditing

4. Critical Production Findings & Recommendations

Finding: Registration Currently Auto-Approves

In `RegistrationScreen.tsx` (lines 119 & 166), when a new food donor or volunteer signs up, the system sets `status: 'Verified'` automatically.

Recommended Production Enhancement: Set `status: 'Pending'` upon registration so that the user appears in the Admin Panel's *Pending Verification Queue*. Full access to posting donations or accepting routes is granted only after the Super Administrator reviews their FSSAI / Aadhaar proof.

Live Real-time Polling Architecture

The Admin Panel utilizes a 3-second background polling cycle against `http://localhost:5000`. If the Express server is stopped, the console gracefully falls back to cached states without crashing, providing high platform fault tolerance.

5. Multi-Service Execution Reference

To execute all three RePlate ecosystem layers concurrently:

```
# 1. Start Central Express Backend (Port 5000)
cd ab
node server.js

# 2. Start RePlate Admin Desktop Console (Port 3000)
cd ab/admin-panel
npm run dev

# 3. Start Food Donor & Volunteer Mobile App (Port 3001)
cd ab/foodsave-master/foodsave-master
npm run dev
```